BREEZE

Your Engineering Partner for AI Hardware

From concept to production -- BOM optimization, DFM, prototyping, and manufacturing. China supply chain expertise, global delivery.

3,000+	12 weeks	1,000
Component suppliers	Concept to prototype	Min pilot run

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No equity. No MOQ surprises. Just engineering.

Building AI Hardware Is Hard.

Finding the right manufacturing partner shouldn't be.

80% of hardware startups fail at the manufacturing stage.

Factories won't take your 1K MOQ

Most factories want 10K+ units. We start at 1,000.

Unpredictable NRE costs

You get a clear, fixed NRE quote before any work starts.

IP leaks and design theft

NDA + IP assignment standard on every engagement. Your designs stay yours.

Language and timezone barriers

Native English + Mandarin team. We work your hours.

No DFM feedback until late

We run DFM checks at the design stage, not after you've committed to tooling.

Certification maze

FCC, CE, UL, SRRC -- we handle the paperwork, testing, and timeline.

Concept to Production in 6 Steps

A clear, predictable path from your idea to manufactured product.

01 Scoping

Share your concept. We assess feasibility, BOM, and ballpark cost.

02 Engineering Design

Schematic, PCB layout, mechanical design, firmware architecture.

03 DFM & Prototyping

Optimize for manufacturability. 3-5 functional prototypes.

04 Tooling & Sourcing

Injection molds, jigs, fixtures. Component sourcing from 3,000+ local suppliers.

05 Pilot Run

50-500 units. Test assembly line, QC process, packaging.

06 Mass Production

Scale to 1K, 10K, 100K+. Ongoing QC, logistics, fulfillment.

12 weeks	$26.40/unit	50-500
Avg concept-to-prototype	AI pendant example	Pilot run units

Cost Reality Check

Real BOM breakdown for a $99 AI wearable pendant (@ 5K units)

$26.40 / unit avg landed cost (incl. assembly + packaging)

Component	Example	Cost Range
SoC / MCU	ESP32-S3 / nRF5340	$1.80 - $4.50
Memory (Flash + RAM)	4MB + 512KB	$0.60 - $1.20
Microphone Array	MEMS x2	$0.80 - $2.00
Speaker / Haptic	Micro speaker + LRA	$0.50 - $1.50
Battery (LiPo)	250mAh pouch cell	$1.20 - $2.50
BLE + Wi-Fi Antenna	Chip antenna / FPC	$0.30 - $0.80
PCB (4-layer HDI)	30x25mm, ENIG finish	$0.80 - $1.50
Enclosure (injection)	PC/ABS, 2-shot mold	$0.60 - $1.20
Connector + Misc	USB-C, pogo pins, ESD	$0.40 - $0.80
PCBA Assembly	SMT + manual + test	$1.50 - $3.00
Packaging	Box, insert, manual, cable	$0.60 - $1.20
TOTAL (@ 5K units)		**$9.10 - $20.20**
Landed Unit Cost*		**$26.40 avg**

** Landed cost includes BOM + PCBA + packaging + testing + 5% yield buffer + freight to US warehouse.*
Excludes NRE (one-time engineering), tooling amortization, and certification fees.

DFM: Top 10 Pitfalls in AI Hardware

Mistakes we've seen (and fixed) across 50+ hardware projects.

1. **Thermal runaway**
   AI inference chips generate 2-5W. Without proper thermal paths, throttling kills performance.

2. **Antenna detuning**
   The human body and enclosure materials shift antenna performance 3-6 dBm. Simulate with real enclosure.

3. **Battery certification gap**
   UN38.3 + IEC 62133 are mandatory for shipping. Budget 6-8 weeks and $3-5K.

4. **Underspec'd power supply**
   Peak current draw during AI inference can be 3x idle. Size your regulator for bursts.

5. **Acoustic cavity design**
   Mic array performance depends on cavity geometry, not just microphone specs.

6. **Injection mold draft angles**
   Below 1.5 degrees on textured surfaces = stuck parts and scratched finish.

7. **Test jig as afterthought**
   Design the test fixture alongside the product. Retrofitting costs 3x and delays 4 weeks.

8. **Connector placement**
   USB-C orientation and position affect waterproofing, drop strength, and user ergonomics.

9. **ESD protection gaps**
   Every external port needs TVS diodes. Skipping them passes lab testing but fails in the field.

10. **Silkscreen and markings**
    FCC ID, CE mark, and recycling symbols need specific sizes and placements per regulation.

Certification Roadmap

What you need, how long it takes, and what it costs.

Certification	Region	Timeline	Budget	Required For
FCC Part 15	United States	4-6 weeks	$3-8K	Any radio transmitter (BLE, Wi-Fi)
CE (RED)	European U...	6-8 weeks	$5-12K	Radio equipment sold in EU/EFTA
UL/IEC 62368	US / Global	8-12 weeks	$8-15K	Audio/video/IT equipment safety
UN38.3	Global	4-6 weeks	$3-5K	Any product with lithium battery
IC (ISED)	Canada	4-6 weeks	$2-4K	Radio devices sold in Canada
SRRC + CCC	China	8-16 weeks	$5-10K	Radio + safety for China market
MIC (Giteki)	Japan	6-10 weeks	$4-8K	Radio equipment sold in Japan

Pro Tip

Start certification pre-compliance testing during the DFM phase.
Finding issues early saves 4-8 weeks and $5-10K in re-spins.
We include pre-compliance checks in our standard engineering process.

Free Tools for AI Hardware Founders

No signup required. Start estimating your project now.

1 BOM Cost Estimator

Input your specs, get a manufacturing cost ballpark in 60 seconds. Covers SoC, memory, sensors, battery, PCB, enclosure, and assembly.

breezehw.com/tools/bom-estimator

2 DFM Checklist

30-point design-for-manufacturability check covering thermal, mechanical, electrical, and certification pitfalls specific to AI hardware.

breezehw.com/tools/dfm-checklist

3 NRE Quote Simulator

Estimate your total development cost: engineering hours, tooling, prototyping, certification, and project management.

breezehw.com/tools/nre-simulator

4 Certification Navigator

FCC, CE, UL, SRRC, UN38.3 -- which do you need? Get a timeline, cost estimate, and testing lab recommendations.

breezehw.com/tools/cert-navigator

Why Breeze?

Four reasons founders choose us over brokers and trading companies.

Whole Product, Not Just Parts

We build the full device -- electronics, enclosure, firmware, packaging. One team, one responsibility, one timeline.

No Equity, No Surprises

NRE + per-unit pricing. Pure service model. We don't take a piece of your company. Period.

Your IP Stays Yours

NDA + IP assignment is standard on every engagement. You receive full source files, schematics, and manufacturing data.

AI Hardware Native

We understand edge AI thermal management, antenna design for wearables, battery optimization, and low-power ML inference.

Breeze vs. the alternatives:

Trading companies add 20-40% margin and zero engineering value.
In-house teams cost $500K+/year and take 6+ months to hire. We're ready now.

Ready to Start?

1. **Try Our Free Tools**
   Get a BOM estimate and DFM checklist at breezehw.com/tools. No signup required.

2. **Book a Scoping Call**
   30 minutes. Share your concept, we assess feasibility and provide a ballpark cost range.

3. **Receive Your NRE Proposal**
   Detailed engineering plan with fixed pricing, timeline, and deliverables. No surprises.

4. **Start Building**
   We assign your dedicated engineering team and begin the 12-week path to prototype.

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NRE + per-unit | No equity | Full IP transfer